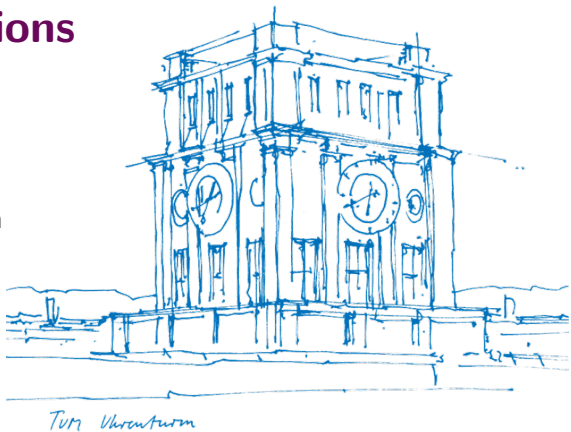
Technical University of Munich

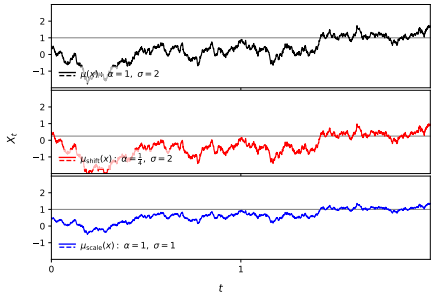
Learning Stationary Diffusions

Fabian Bleile, Sarah Lumpp, Mathias Drton

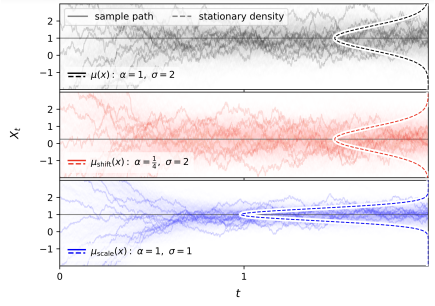


Stationary Diffusions

- (time-homogeneous) SDE: $dX_t = \underbrace{b(X_t)}_{\text{'drift' / deterministic}} dt + \underbrace{\sigma(X_t)}_{\text{'diffusion'/stochastic}} dB_t$



sample paths, hitting times,
stationary distribution



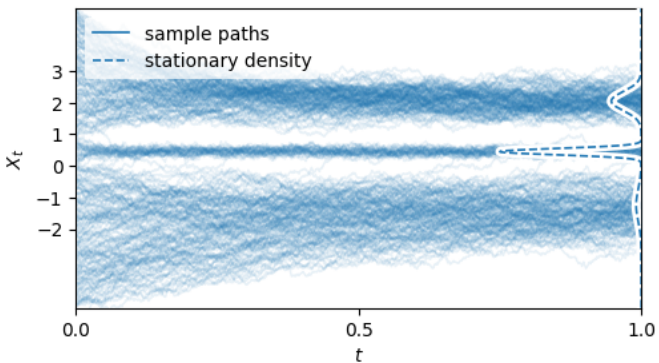
Generative modelling, MCMC sampling,
Graphical / Causal Models

Example

- Langevin Dynamics:

$$dX_t = \nabla \log \mu(X_t)\, dt + \sqrt{2} I_d\, dB_t$$

- In general: stationary distribution exists uniquely
  1. Regularity condition
  2. Non-dissipative condition



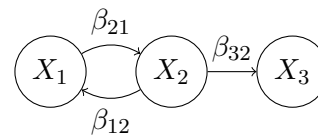
Stationary Diffusion as a Graphical Model[1]

$$dX_t = \begin{pmatrix} -a_1 & \beta_{12} & 0 \\ \beta_{21} & -a_2 & 0 \\ 0 & \beta_{32} & -a_3 \end{pmatrix} X_t \, dt + I_3 \, dB_t$$

Stationary distribution

$$X \sim \mathcal{N}(0, \Sigma)$$

Graph

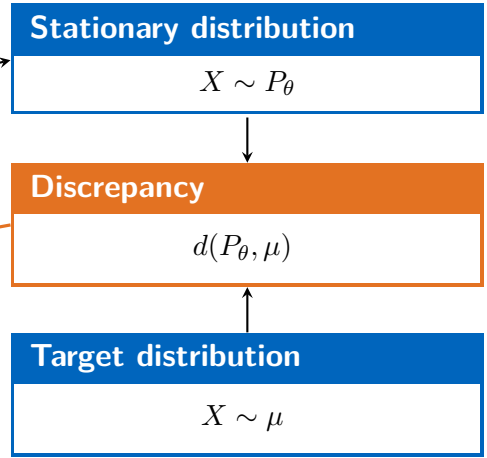


- Most rigorous work for linear SDEs:
  - Identifiability $\mu \mapsto (b, \sigma)$ [Amé+25], [Det+23]
  - Conditional Independence Structures [Boe+24]

[1] [HS14], [VH20]

Learning Stationary Diffusions

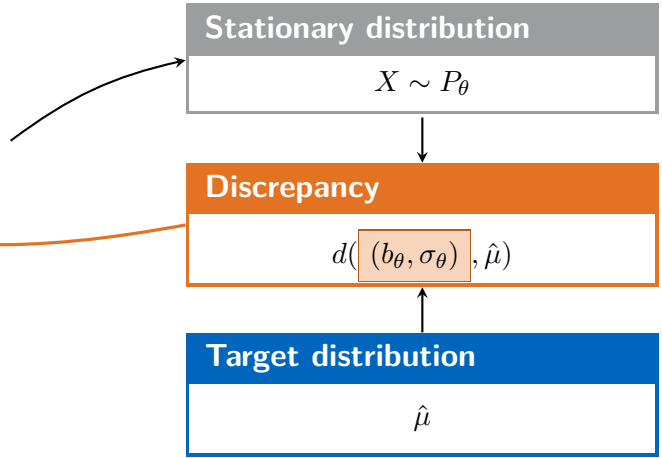
$$dX_t = \underbrace{b_\theta(X_t)}_{\text{parametrized}}\, dt + \underbrace{\sigma_\theta(X_t)}_{\text{parametrized}}\, dB_t$$



- Common settings:
  1. Sampling from P_θ and μ is feasible $\implies$ two-sample discrepancies
  2. Normalized density of μ is known $\implies$ e.g. Langevin diffusion

Learning Stationary Diffusions

$$dX_t = \underbrace{b_\theta(X_t)}_{\text{parametrized}}\, dt + \underbrace{\sigma_\theta(X_t)}_{\text{parametrized}}\, dB_t$$



- Our setting:
  1. Samples from P_θ too expensive
  2. Normalized density of μ not accessible

Step 1: Fokker-Planck Equation

- PDE that describes density in time of stochastic process given by $dX_t = b(X_t)dt + \sigma(X_t)dB_t$

$$\begin{aligned} \partial_t p(t,x) &= -\nabla_x \cdot \left(b(x)p(t,x) - \frac{1}{2}\sum_j \partial_j \left((\sigma\sigma^T)_{\cdot j}(x)p(t,x) \right) \right) \\ &= \mathcal{L}^\star p(t,x). \end{aligned} \tag{1}$$

- **Stationary** if $\partial_t p(t,x) = 0$

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- **Stationary** if $\partial_t p(t,x) = 0$
- For $\mu \in C^2$, testing $\mathcal{L}^\star \mu = 0$ is feasible
- For empirical distribution $\hat{\mu}$, $\mathcal{L}^\star \hat{\mu} = 0$ not well-defined
- Idea:

  Kernel smoothing of $L^\star \hat{\mu}$

Step 2: Kernel Mean-Embedding

- Kernel mean-embedding for $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ into RKHS $\mathcal{H}_k$

$$\nu \mapsto \int_{\mathbb{R}^d} k(\cdot, x) d\nu(x) \in \mathcal{H}_k \tag{2}$$

- Injective for universal kernel k and on σ-finite measures

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$$\mathcal{L}^\star \mu(x) = 0 \iff \left\| \int_{\mathbb{R}^d} k(\cdot, x) \mathcal{L}^\star \mu(x) dx \right\|_{\mathcal{H}}^2 = 0 \tag{3}$$

- Kernel Deviation from Stationarity (Lorch, Krause, and Schölkopf [LKS24])

$$\begin{aligned} \left\| \int_{\mathbb{R}^d} k(\cdot, x) \mathcal{L}^\star \mu(x) dx \right\|_{\mathcal{H}}^2 &= \| \mathbb{E}_\mu \left[\mathcal{L} k(\cdot, x) \right] \|_{\mathcal{H}}^2 \qquad \text{(Integration by Parts)} \\ &= \mathbb{E}_{\mu \otimes \mu} \left[\mathcal{L}_1 \mathcal{L}_2 k(x, y) \right] \end{aligned}$$

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Stein-Type Kernel Deviation from Stationarity

- Fokker-Planck Equation

$$\begin{aligned}\partial_t p(t,x) &= -\nabla_x \cdot \underbrace{\left(b(x)p(t,x) - \frac{1}{2}\sum_j \partial_j\left((\sigma\sigma^T)_{\cdot j}(x)p(t,x)\right)\right)}_{=:J \text{ (probability flux / current)}} \\ &= \mathcal{L}^\star p(t,x) \\ &= -\nabla_x \cdot \boxed{(\mathcal{S}^\star p(t,x))}.\end{aligned} \tag{5}$$

- Detailed Balance condition: $J = \mathcal{S}^\star p(x) = 0$

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- Detailed Balance condition: $J = \mathcal{S}^\star p(x) = 0$
- **Stationary** and **Reversible**
- Similar as before

$$\mathrm{SKDS}(\boxed{S}, \mu) =: \mathbb{E}_{\mu\otimes\mu}\left[\mathcal{S}_1\mathcal{S}_2 k(x,y)\right] \tag{6}$$

Comparison

KDS	Stein-type KDS
$\mathrm{KDS}(\mathcal{L}, \mu) := \mathbb{E}_{\mu\otimes\mu}\left[\mathcal{L}_1\mathcal{L}_2 k(x,y)\right]$	$\mathrm{SKDS}(\mathcal{S}, \mu) =: \mathbb{E}_{\mu\otimes\mu}\left[\mathcal{S}_1\mathcal{S}_2 k(x,y)\right]$
$\mathcal{L}h = \langle b, \nabla_x h\rangle + \frac{1}{2}\operatorname{tr}\left(\sigma\sigma^T \nabla_x \nabla_x h\right)$ $h : \mathbb{R}^d \to \mathbb{R}$	$\mathcal{S}g = \langle b, g\rangle + \frac{1}{2}\operatorname{tr}\left(\sigma\sigma^T \nabla_x g\right)$ $g : \mathbb{R}^d \to \mathbb{R}^d;$
$\mathcal{L}_1\mathcal{L}_2 k(x,y)$: 4th order kernel evaluations	$\mathcal{S}_1\mathcal{S}_2 k(x,y) I_d$: 2nd order kernel evaluations
characterizes μ-stat. diffusion	characterizes μ-stat. diffusion + reversibility
Samples from P_θ too expensive ✓ and only samples of μ accessible ✓	

Interesting Results and Further Work

$$\mathrm{SKDS}(\mathcal{S}, \mu) =: \mathbb{E}_{\mu\otimes\mu}\left[\mathcal{S}_1\mathcal{S}_2 k(x, y)\right]$$

- If $S^\star p = 0$, then $\mathrm{SKDS}(\mathcal{S}, \mu) \leq C\mathcal{W}_2(p, \mu) +$ irreversible part
- SKDS **convex** under (linear parametrization)
- Further Work: goodness-of-fit test with appropriate kernel choice

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[Amé+25] C. Améndola et al. "Structural Identifiability of Graphical Continuous Lyapunov Models". In: *arXiv preprint arXiv:2510.04985* (2025).

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[Det+23] P. Dettling et al. "Identifiability in continuous Lyapunov models". In: *SIAM Journal on Matrix Analysis and Applications* 44.4 (2023), pp. 1799–1821.

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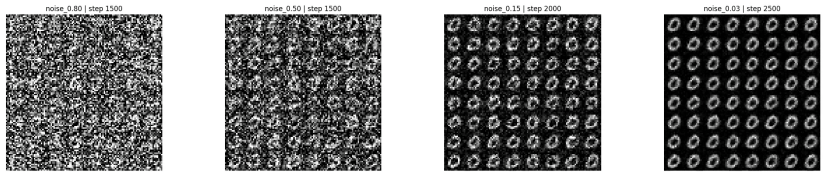
[LKS24] L. Lorch, A. Krause, and B. Schölkopf. "Causal Modeling with Stationary Diffusions". In: *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR. 2024, pp. 1927–1935.

[VH20] G. Varando and N. R. Hansen. "Graphical continuous Lyapunov models". In: *Conference on Uncertainty in Artificial Intelligence*. Pmlr. 2020, pp. 989–998.

Generative Modelling with SKDS

- Playground:

Figure 1 Scan QR code for video.



Kernel Deviation from Stationarity [LKS24]

- Definition: Diffusions with μ as stationary distribution
- Characterization: μ stationary solution of SDE if and only if

$$\mathbb{E}_{X\sim\mu}[\mathcal{L}h(X)] = 0, \quad \text{for all } h \in C_c^\infty(\mathbb{R}^d) \tag{7}$$

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- Infinitesimal generator $\mathcal{A}$,

$$\begin{aligned} \mathcal{A}h(x) &= \lim_{t \downarrow 0} \frac{1}{t} (\mathbb{E}_{X_t}[h(X_t) \mid X_0 = x] - h(x)) \\ &= \langle b(x), \nabla_x h(x) \rangle + \frac{1}{2} \operatorname{tr}\left(\sigma(x)\sigma(x)^T \nabla_x \nabla_x h(x)\right) \end{aligned}$$

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$$KDS(\mathcal{L}, \mu) = \sup_{h \in \mathcal{H}} \mathbb{E}_{X \sim \mu}[\mathcal{L}h(X)] = \underbrace{\mathbb{E}_{X \sim \mu}\mathbb{E}_{Y \sim \mu}\left[\mathcal{L}_1 \mathcal{L}_2 k(X, Y)\right]}_{\text{closed form solution}} \tag{8}$$

This work

- Characterization: μ is reversible stationary solution of SDE if and only if

$$d(\mathcal{S}, \mu) = \sup_{g \in C_c^\infty(\mathbb{R}^d, \mathbb{R}^d)} \mathbb{E}_{X \sim \mu}[\mathcal{S} g(X)] = 0 \tag{9}$$

- Operator $\mathcal{S}$,

$$\mathcal{S} g(x) = \langle b(x), g(x) \rangle + \frac{1}{2} \operatorname{tr}\left(\sigma(x)\sigma(x)^T \nabla_x g(x)\right)$$

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- Why is (9) a characterization?
- How to arrive at a computable thing? (10)

Link to Stein-Discrepancy – Sketch

- Integral probability metric:

$$P = Q \iff \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{X \sim P} f(X) - \mathbb{E}_{Y \sim Q} f(Y) \right| = 0 \tag{11}$$

- Stein-Discrepancy for operator $\mathcal{T}_Q$

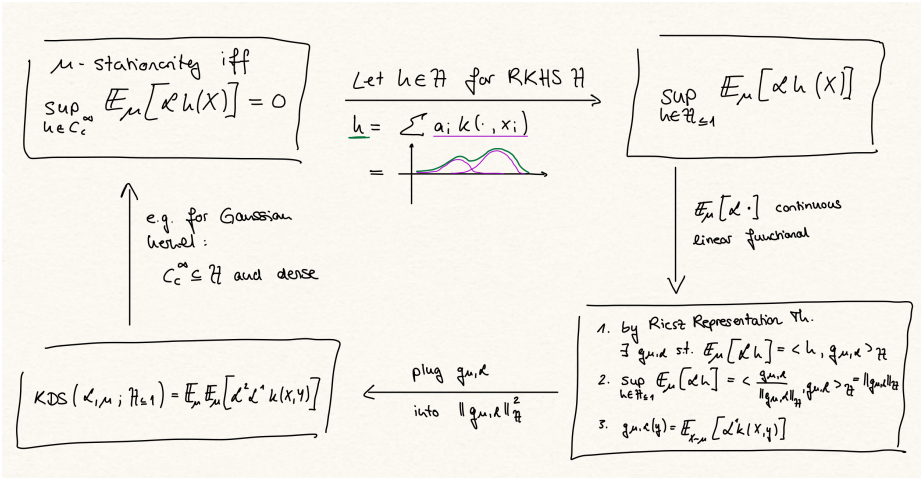
$$P = Q \iff \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{X \sim P} \mathcal{T}_Q f(X) - \underbrace{\mathbb{E}_{Y \sim Q} \mathcal{T}_Q f(Y)}_{=0} \right| = 0 \tag{12}$$

- 'Implicit' Stein-discrepancy for operator $\mathcal{S}$

$$P = Q \iff \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{X \sim P} \mathcal{L} f(X) - \underbrace{\mathbb{E}_{Y \sim Q} \mathcal{L} f(Y)}_{=0 \text{ if } Q \text{ stationary solution}} \right| = 0 \tag{13}$$

Kernel Deviation from Stationarity

- [LKS24], *Causal Modeling with Stationary Diffusions*



Why *learning* a stationary diffusion?

- Definition: Given samples from μ, return μ-targeted stationary diffusion (drift b, diffusion σ)

Why *learning* a stationary diffusion?

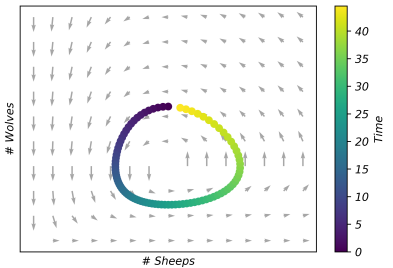
- Definition: Given samples from μ, return μ-targeted stationary diffusion (drift b, diffusion σ)
- For GenAI: $\nabla \log(\mu(x))$ works great! No need?
- For MCMC samplers: Design drift to have favorable convergence and mixing?
- Our interest: Graphical / Causal models, e.g. for systems with feedback loops
  - ☐ Model joint distributions
  - ☐ Resolve feedback-loops in time
  - ☐ Meaningful drift-sparsity and interventions

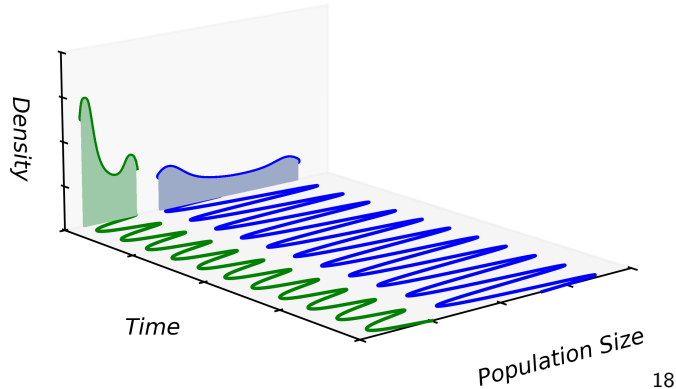
Example: Predator-Prey Model

- Wolves W_t $\longleftrightarrow$ Sheep S_t
- Model: $d\begin{pmatrix} S_t \\ W_t \end{pmatrix} = b\begin{pmatrix} S_t \\ W_t \end{pmatrix} dt + \sigma \begin{pmatrix} S_t \\ W_t \end{pmatrix} dB_t = \begin{pmatrix} \alpha S_t - \beta S_t W_t \\ \delta S_t W_t - \gamma W_t \end{pmatrix} dt$

Example: Predator-Prey Model

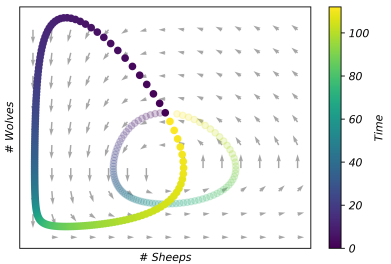
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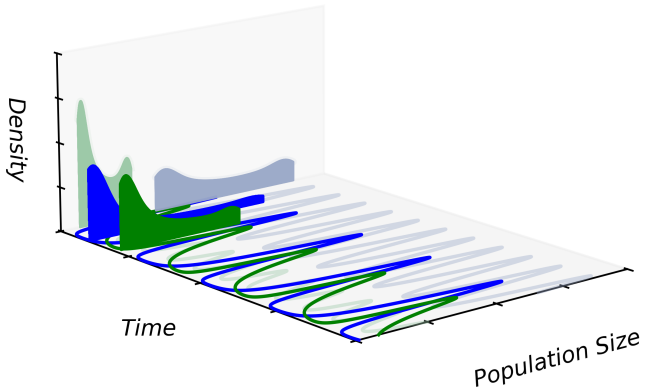




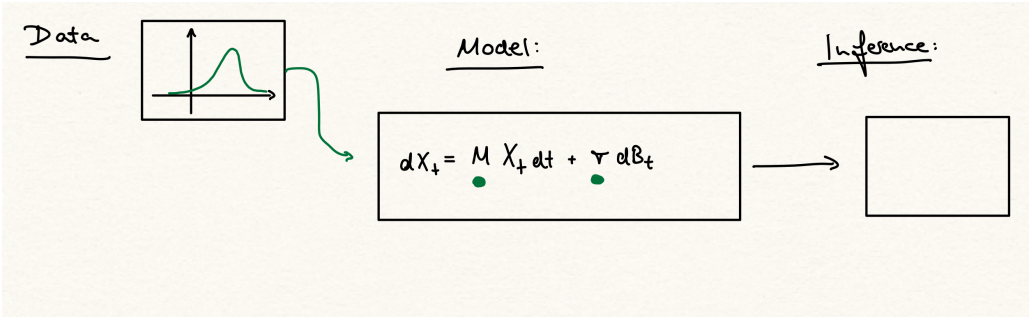
Intervention: Predator-Prey Model

- Intervention ϕ: $dX_t = b_\phi(X_t)dt + \sigma_\phi(X_t)dB_t$
- Decreased natural death rate γ of wolves

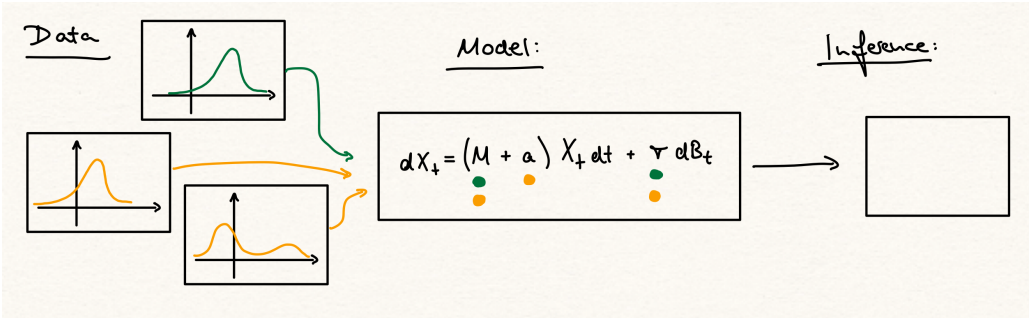




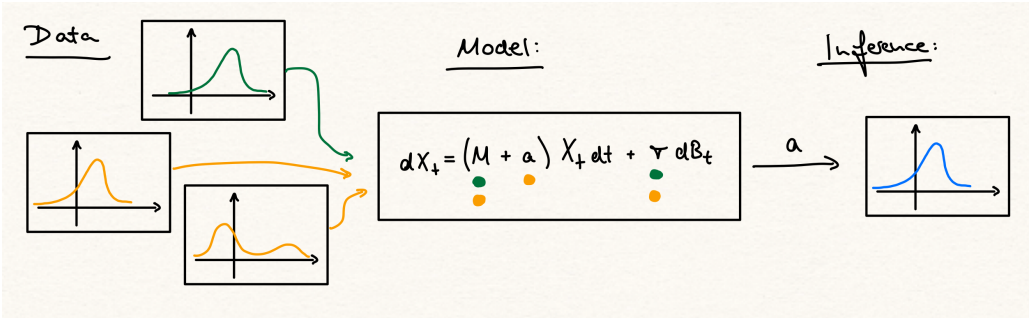
Learning Stationary Diffusions with Interventional Data

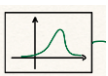


Learning Stationary Diffusions with Interventional Data



Learning Stationary Diffusions with Interventional Data



- Target $\mu =$ 
- How to 'learn' parameters, such that SDE has targeted stationary solution μ?

Stein Discrepancies

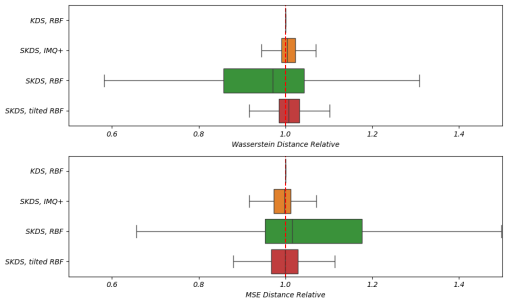
- Integral probability metric:

$$d_{\mathcal{F}}(Q, P) := \sup_{f \in \mathcal{F}} |\mathbb{E}_{X \sim Q}[f(X)] - \mathbb{E}_{X \sim P}[f(X)]|$$

- For well chosen $\mathcal{F}$:
  - $d_{\mathcal{F}}(Q, P) = 0$ iff $Q = P$
  - $d_{\mathcal{F}}(Q_n, P) \to 0$ iff $Q_n \to P$ (weak convergence)
- Stein discrepancy:
  1. $\mathcal{F} = \mathcal{T}_P[\mathcal{G}]$ such that $\mathbb{E}_{X \sim P}[f(X)] = 0$ for all $f \in \mathcal{T}_P[\mathcal{G}]$
  2. $d_{\mathcal{T}_P[\mathcal{G}]}(Q, P) = \sup_{g \in \mathcal{G}} |\mathbb{E}_{X \sim Q}[\mathcal{T}_P g(X)]|$
- *diffusion Stein operator*: $\mathcal{T}_P g = \frac{1}{p}\langle \nabla_x, pmg \rangle$ (divergence: $\langle \nabla_x, \cdot \rangle$)
- Problem: No access to p during training. Instead we have drift b and σ.

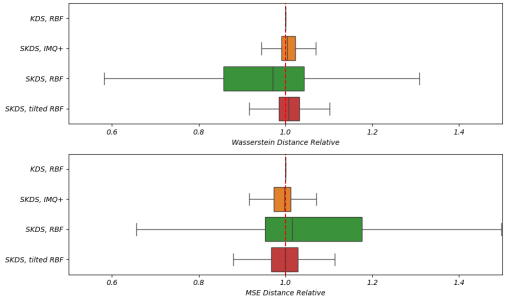
Experiments

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- Stein-type KDS inherits properties of KDS
  - Characterizes stationarity
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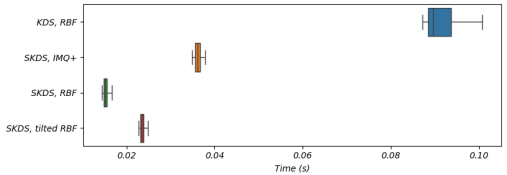


Figure 2 Faster computation time, between 2.6 to 6 times faster. S-operator involves only first-order derivatives

Gradient descent on SKDS

- Model: $dX_t = -4(X_t - a)dt + cdB_t$
  - Ground-truth: $a = 1,\ c = 2$
  - Wrong shift: $a = 1/4,\ c = 2$
  - Wrong scale: $a = 1,\ c = 1$

